

Assignment 2 - Data Structures for Spatial Computing

Submission Date: Tuesday, September 15, 2026 (by 11.59pm on Canvas)

This assignment utilizes concepts introduced in the Week 3 notebook, 3_01_Data_Structures.ipynb. You will use lists, tuples, dictionaries, and strings to organize survey routes and point-of-interest records for spatial computing.

Complete the work in a new Google Colab notebook. Put your name, course number, and assignment title at the top; label every part clearly; run every code cell; and leave all output visible. Use descriptive snake_case names and include units where appropriate.

Table 1 outlines the required assignment components.

Component	Requirements	Points
1. Index and update a route	Create route = ["Depot", "Library", "Park", "Museum", "Station"]. Print the first stop, the last stop using a negative index, and the first three stops using a slice. Before running the slice, predict its output. Replace "Library" with "Clinic", append "River pier", and remove "Museum". Print the revised route, its length, and whether "Park" is present.	10
2. Copy and order lists	Create original_route = ["Wetland", "Forest", "Meadow"]. Create alias_route by assignment and revised_route using .copy(). Append "Stream" through alias_route and "Hill" to revised_route. Print all three lists and explain which changes are shared and why. Use sorted() to create an alphabetically ordered version of revised_route; print both lists. Then use .sort() on revised_route and print it. Explain how sorted() and .sort() differ.	15
3. Store coordinate tuples	Create station_coordinates = (38.84, -77.31). Unpack it into latitude and longitude, and print both with labels and units. Print the tuple length. Create a new tuple with latitude 38.85 and the original longitude; print both tuples. Explain why assigning directly to station_coordinates[0] would fail, without executing that invalid assignment. Create a single-element tuple containing the latitude and print its type.	15
4. Manage a site dictionary	Create a dictionary named survey_site with name "Pond station", city "Arlington", and coordinates (38.83, -77.30). Print its name and coordinates by key. Change the city to "Fairfax". Use .get() to retrieve opening_hours with default "Not recorded"; then add opening_hours with value "09:00-17:00" and repeat the lookup. Remove city using .pop() and print the removed value, the final dictionary, and its keys. Explain why the two lookups differ.	15
5. Compare survey coverage	Create team_a_visits = ["Pond", "Forest", "Pond", "Meadow"] and team_b_visits = ["Forest", "Stream", "Hill"]. Convert each to a set. Print the length of team_a_visits and its set, explaining the difference. Predict the shared sites, then calculate and print the intersection, union, difference (A minus B), and symmetric difference. Explain what each result means for survey coverage. Use .discard() to remove "Quarry" from team A's set and explain why no error occurs.	15
6. Clean spatial text	Create raw_address = " 120 River Road, Fairfax, VA ". Use .strip() to create clean_address, then split it at each comma followed by a space. Print the resulting list, and both the city and state using indexing. Join the parts with " " and print the result. Create a lowercase version of clean_address with spaces replaced by underscores. Print raw_address and explain why the original string remains unchanged.	10

Component	Requirements	Points
7. Build a nested spatial dataset	Create a list of three point-of-interest dictionaries. Each must contain name and category strings, a coordinates tuple, and a list of services offered (e.g., “food”, “dry cleaning”, etc.). Give the first two points at least one shared service. Print the last point dictionary via negative indexing. Add a service to the first point and print the services shared by the first two points. Make a shallow copy of the list, change the first dictionary's category through the copy, and print that category through both lists. Explain the shared change and also why lists, dictionaries and tuples may be suitable data structures for items in this dataset.	20

Table 1 Assignment components and requirements.

Submit the completed .ipynb file online. Before submitting, restart the runtime and run the notebook from top to bottom. Confirm that all seven parts, requested outputs, predictions, and written explanations are present, and no errors remain. Use Python operations rather than manually typing results. Coordinate tuples store latitude followed by longitude in decimal degrees; fictional locations are acceptable.

Also submit a .PDF file of your notebook (or a .PDF of screenshots), for easy review.

Table 2 outlines the assessment criteria. Component points total 100; the criteria describe how the work is assessed across those components.

Table 2 Assessment criteria.

Criterion	Evidence	Weight
Correctness / completeness	All seven parts are present; indexing, updates, set operations, text processing, and nested access produce correct results.	40%
Data structure selection / use	Appropriate structures and methods are used; code demonstrates ordering, mutability, uniqueness, key-based access, and copying.	25%
Documentation / readability	Names preferably follow snake_case, units and labels are clear, and comments, predictions, and written explanations are useful.	15%
Testing / output	All requested operations are run. Visible outputs show before-and-after states where requested and support the explanations.	15%
Organization / submission	The notebook is clearly labeled, logically ordered, complete, and submitted in the required formats.	5%

On this particular assignment, you are not allowed to use AI, otherwise it defeats the point of trying to learn this content. All the functions you need should be in the notebook, and you are allowed to read the Python documentation for extra background. The knowledge from this assignment is cumulative and necessary to succeed in later weeks of the course (and especially in the pencil and paper exam where you cannot use AI). Undisclosed AI-generated work, fabricated material, or code that the student cannot explain may constitute an Honor Code violation.